Seeing Is Not Believing: Counterfactual Lesion Erasure Teaches a Small Medical Vision–Language Agent to Report Only What the Image Shows

Bhaskar Gurram
Independent researcher github.com/bhaskargurram-ai/medscope

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Draft for expert review. The confirmatory numbers come from the pre-registered v4 runs (notebooks 23–25); the post-hoc pneumothorax analysis (notebooks 26–27), the post-review checks (notebooks 29, 31, 32) and the pre-registered mechanism checks (notebooks 33–34) are labelled as such. Every number is regenerated from stored per-view transcripts. Items in orange brackets are open: the reader study (Section 7), a few values to be pulled from the stored runs, and a final check of every reference. Not for reader-study participants until they have submitted their answers.

Abstract

Background. Medical vision–language models (VLMs) often report a finding because the question names it; standard benchmarks miss this.
Methods. From public segmentation datasets we built counterfactual views without manual annotation: the lesion erased by inpainting, a same-shape control edit elsewhere, and unedited lesion-free images. The *evidence sensitivity gap* (ESG) is the rate of "absent" answers after erasure minus the rate after the control edit; it is zero for a model that ignores the image or reacts to both edits alike. MedScope-4B (Qwen3.5-4B, LoRA, image tools) was trained on label-derived trajectories, selected by a pre-registered rule and tested once against nine frontier and six open VLMs, then compared, under pre-registered rules, with an unmatched control edit and with removal-based preference training (DPO).
Results. On 507 internal cases across five imaging types, MedScope-4B reached ESG 0.758 and claimed an erased lesion in 9.6% of cases, against 0.27–0.47 and 41–62% for the frontier models (all $p < 0.001$), without significant loss of accuracy on real lesions. Without counterfactual views ESG fell to 0.279 ± 0.107 (three seeds); the effect replicated on MedGemma 1.5 4B and on two unseen inpainting tools. At four external centres the advantage held but frontier models detected more real lesions, and on a fresh external test MedScope-4B missed every pneumothorax. Matching the control edit was not shown to matter; removal-based DPO reached the recipe's ESG on new data with more erased-lesion claims and more sensitivity.
Conclusions. Training on erased and edited images gives a small model an evidence-gated disposition that prompting does not give frontier models; it does not improve perception, and rare critical findings need explicit coverage.

Keywords: medical vision–language models; hallucination; counterfactual data; evidence sensitivity; inpainting; pre-registration; agentic AI

Highlights

- Erasing a lesion, with a matched control edit, tests if answers follow the image.
- A 4B agent claims erased lesions in 9.6% of cases; nine frontier models, 41–62%.
- Without counterfactual views the rate is 68% (three seeds); replicated on MedGemma.
- The gain holds on two inpainting tools never used in training and under six prompts.
- Pre-registered checks: control matching not shown to matter; DPO ties on ESG.
- A pre-registered fresh test exposed a complete pneumothorax miss, linked to coverage.

1 Introduction

A radiologist asked "Is there a polyp?" looks for one. A vision–language model asked the same question may answer "Yes, a sessile polyp is visible" whether or not the image contains one. In the image–text data these models learn from, a question that names a finding is itself strong evidence that the finding is present, and a model can learn to answer from the question rather than from the pixels. For a clinical assistant this is the most dangerous kind of error: a confident, plausible report of something that is not there.

Medical VLMs are now built by instruction-tuning general models on biomedical data [1, 2, 3, 4, 5] and are evaluated mainly on VQA-RAD, SLAKE and PathVQA [6, 7, 8]. In these benchmarks the presupposition of a closed question is usually true, so a model that leans on the question scores well. Hallucination benchmarks [9, 11, 12, 13, 14] probe absent findings, but they change the question or the answer while the image stays the same, so they cannot tell whether a correct "No" came from the image or from a learned prior about the question.

Testing whether an answer depends on the image needs image pairs that differ only in the finding. Such pairs cannot be acquired clinically for most findings, and expert checking of synthetic pairs at scale is costly. We build them automatically from public segmentation datasets by erasing the expert-masked lesion with learned inpainting [15]. The obvious confound is that a model could learn to detect *that an image was edited* rather than *whether the lesion is there*. Every lesion-erased view therefore has a twin in which a region of exactly the same shape is erased elsewhere with the same tool, leaving the lesion intact. A model that responds to the editing signature alike in both views cannot separate them; treating them differently is consistent with reading the lesion, although anatomy-dependent traces of the erasure can also contribute (Section 5.4).

We use these views twice: to measure evidence sensitivity, and to train it. A 4-billion-parameter agent that can zoom and window the image is trained on trajectories derived from dataset labels alone, selected by a rule committed before any of its data existed, and tested once against nine frontier and six open models through an identical agent loop.

Contributions.

1. **Counterfactual evidence views with an edit-matched control**, built and quality-checked automatically across five imaging types, with no human annotation (Section 3.2).
2. **The evidence sensitivity gap (ESG)**, zero for every constant policy and, in expectation, for every policy that ignores the image or responds to the two edits alike, reported with false findings on erased lesions and on unedited healthy images, grounding, and a clinical cost-matrix utility (Section 3.8).
3. **MedScope-4B**, selected by a pre-registered rule, which claims erased findings at 4–6 times lower rates than nine frontier models, with no significant difference in accuracy on real lesions on the internal test, robustly to prompting, question wording and unseen inpainting tools (Sections 5.1–5.5).
4. **Evidence on what produces the behaviour**: a three-seed ablation of the counterfactual views, an arm matched for the share of "absent" answers, an erasures-only arm, a 2×2 factorial of control edits and insertions, a second base model, and two pre-registered checks whose rules were committed before the arms were trained: whether matching the control edit matters (it was not shown to), and how the recipe compares with removal-based preference training, the closest existing method (Sections 5.3, 3.7).
5. **A like-for-like baseline and its failure profile**: removal-based DPO adapted to the same cases and base model reaches the recipe's evidence sensitivity on new data while claiming more erased lesions and keeping more sensitivity; the two methods are complementary rather than ranked (Section 5.3).
6. **A transparent account of failure**: at new centres frontier models detect more real lesions, and a confirmatory fresh test revealed a complete miss of pneumothorax, which a post-hoc analysis links to training coverage (Sections 5.6, 6).

Table 1: Closest related work. "Image changed": the probe alters the image rather than the question. "Control edit": an edit elsewhere that leaves the finding intact. "Trains": the method is used to train, not only to measure.

Work	Image changed	Control edit	Lesion-free symmetry	Trains	Medical	Compared here
POPE [9], CARES [11], MedVH [13], MedHEval [14]	no	–	–	no	partly	–
Causal VQA edits [18]	yes	unmatched	no	no	no	unmatched arm (§5.3)
Object removal + preference optimisation [10]	yes	no	no	yes	no	removal-based DPO arm (§5.3)
Counter-evidence and counterfactual grounding detectors [16, 17]	at inference	no	no	no	partly	–
This work	yes	matched and unmatched	yes	yes	yes	–

2 Related work

Medical vision–language models. General medical assistants are built by instruction-tuning general VLMs on biomedical image–text data: LLaVA-Med [1], HuatuoGPT-Vision [2], MedGemma [3, 4] and Lingshu [5]. They are evaluated mostly on VQA-RAD [6], SLAKE [7] and PathVQA [8], where the asked finding is usually present and accuracy rewards agreement with the question's presupposition.

Hallucination benchmarks and detectors. POPE [9] probes object hallucination with questions about absent objects, and He et al. [10] remove objects from natural images to test whether a VLM still "sees" them, then mitigate the failure with preference optimisation. Medical benchmarks — CARES [11], Med-HallMark [12], MedVH [13] and MedHEval [14] — curate or perturb questions and answers. Two recent training-free detectors use counterfactual probes at inference: counter-evidence verification [16] blacks out the region supporting a claim and checks whether the claim persists, and counterfactual visual grounding uncertainty [17] contrasts grounding on factual and perturbed inputs. Our work differs from the detectors in that we change the *image* while holding the question fixed and use the views to *train* a model, not only to audit one. It differs from removal-based evaluation and mitigation [10] in the control edit, in the use of unedited lesion-free images scored identically to erasures, in the agent protocol with abstention and clarification, and in the medical setting, where a reference-standard mask defines what is removed. Two pre-registered checks (Section 5.3) bound the difference: matching the control edit's shape and tool to the erasure was not shown to change what the model learns, and a removal-based preference baseline [10] adapted to the same cases reaches the same evidence sensitivity on new data. The contribution relative to that line of work is therefore the measurement (an evidence-sensitivity metric with a built-in control and a cost-matrix utility), the medical construction and its validation, and the characterisation of what each training method gets wrong, not a superior training principle. Table 1 places the closest methods side by side.

Counterfactual image editing. Agarwal et al. [18] edited natural images for VQA, removing objects so that the answer should change (covariant edits) or irrelevant objects so that it should not (invariant edits), and showed that VQA models rely on spurious correlations. Our target and random removals are the medical analogue, with two additions: the invariant edit uses the same tool and the *same shape* as the covariant one, and abstention and clarification are scored by a cost matrix. We do not claim the editing idea as new; the contribution is the matched control, the natural-negative symmetry and the training recipe, and Section 5.3 measures what each component adds in a factorial ablation. Learned inpainting such as LaMa [15] makes plausible fills of large masks possible; classical inpainting [19], Poisson blending [20] and diffusion inpainting [21] provide independent tools for testing whether a model learned one tool's traces.

Tool-using and reinforcement-trained agents. Recent agents zoom or crop to "think with images" and are trained with reinforcement learning [22, 23]; in medicine, MMedAgent [24] learns to call tools and MMedAgent-RL [25] optimises multi-agent collaboration. Our agent uses two simple tools. On-policy RL (GRPO [26] with DAPO-style dynamic sampling [27]) gave no signal in pilot runs, so we use rejection-sampling fine-tuning [28, 29]. Selective prediction is summarised by the area under the risk–coverage curve [30]. We follow the CLAIM 2024 checklist [31] (Appendix C) and note the items of TRIPOD-LLM [32] and STARD-AI [33] that apply.

3 Methods

3.1 Task and agent protocol

An episode presents one image and one question whose answer depends on a lesion. Each turn the model writes brief reasoning followed by exactly one tagged JSON action (prompt in Appendix A): `zoom` (a box on the original image, integer coordinates 0–1000, returning the cropped region) or `window` (centre and width on a 0–255 scale, returning the contrast-adjusted image), at most three calls per episode; `answer`, with an optional evidence box and a stated confidence; `refer` (usable image, cannot decide); or `clarify` (unusable image or wrong anatomy). When the asked finding is not visible the model must say so ("No" to "Is there…?", "Yes" to "Is …healthy?", "No abnormality" to an open question). Every model in this paper receives the same prompt, tools and turn structure.

3.2 Counterfactual views

For every case (image, expert lesion mask, question whose answer depends on the lesion) we build four views; unedited lesion-free images from the same sources form a fifth kind (Figure 1; real test views in Figure 2).

Original. Unchanged. Expected: the correct answer with a box on the lesion.
Target removal. The lesion mask, dilated by 5 pixels, is inpainted. Expected: the answer stating the finding is absent.
Random removal (control). The same dilated shape, moved to a random foreground position that does not touch the dilated lesion, inpainted with the same tool. No lesion pixel changes. Expected: the original answer.
Degraded. Gaussian blur ($\sigma = \max(h,w)/40$), Gaussian noise ($\sigma = 60$), 8× down- and up-sampling, JPEG compression, motion blur, or an image of a different imaging type. Expected: `clarify`.
Natural negative. An unedited lesion-free image with the dataset's label. Expected: the absent answer, with exactly the reward of target removal.

Two symmetries make the design work. Target removal and natural negatives share expected answer and reward, so a policy of answering "absent" whenever an image looks edited earns nothing over the benchmark as a whole. Target and random removal share tool and shape, so a policy that responds to the editing signature alike in both views scores ESG 0. The symmetry is not a proof of immunity: erasing a lesion from its anatomical context can leave traces that erasing elsewhere does not, and Section 5.4 tests this directly. Lesions under 20 pixels or over 20% of the image are excluded. Training additionally uses lesion-insertion views (a real lesion moved into a healthy training image; expected: present) and a second erasure tool, tissue transplant (tissue from elsewhere in the same image, Poisson-blended [20]), for half of the cases.

3.3 Automated quality control

No human checks any view. A case is kept only if (i) the lesion no longer stands out after target removal: with contrast defined as the absolute difference between the mean grey level inside the mask and in a ring 6–14 pixels around it, in units of the ring's standard deviation, the contrast after erasure must be at most $\max(1.0,\ 0.35 \times \text{contrast before})$, and a trained edit detector must score the erased region below 0.5; (ii) no

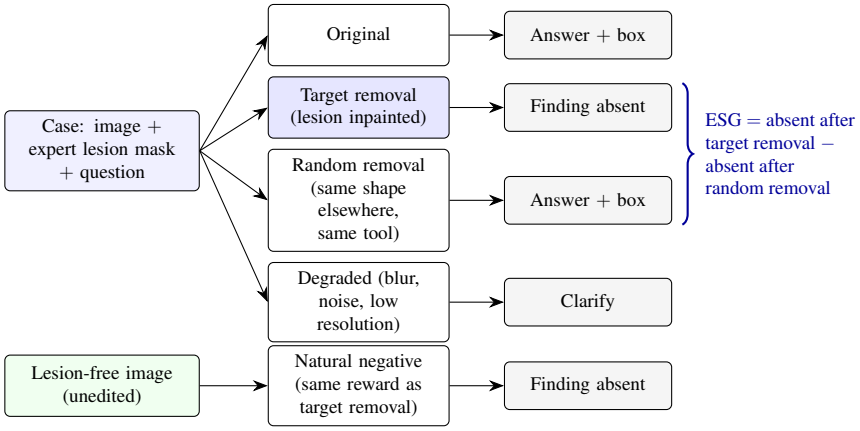


Figure 1: Views built from each case and the expected action. Target removal and the random-removal control share the editing tool and the edited shape, so a model that responds only to the shared editing signature treats them alike; ESG measures the difference. Natural negatives share target removal's expected answer and reward, so answering "absent" to anything that looks edited earns nothing overall.

Table 2: Cost matrix (reward per view). Target removal and natural negatives are scored identically by design.

View	Correct / absent answer	Wrong, or claims finding	Refer	Clarify
Original, random removal	$+1.0$	-2.0 (-4.0 critical)	-0.3	-0.3
Target removal, natural negative	$+1.0$	-2.0	$+0.5$	-0.5
Degraded	$+0.25$	-2.0	0.0	$+0.5$

lesion pixel changed in random removal, and (iii) no view is a near-copy of a training image (perceptual hash, confirmed by pixel correlation). Splits are made over patients, or over groups of near-identical images where patient identifiers are missing; SLAKE's official split leaks neighbouring slices (4 of 63 validation and test images had a near-copy in training), which group splitting removes. A texture classifier trained to separate target from random edits drives adversarial filtering until its AUC is at most 0.6 (test set: 0.56). A hand-crafted shortcut classifier on the v4 training views reached AUC 0.59, so no further filtering was needed.

3.4 Reward and net utility

Each decision is scored with the cost matrix in Table 2. It is the selection reward during training and the *net utility* metric at test. The matrix is an assumption chosen before any experiment to penalise false findings and missed critical findings most; it was not elicited from clinicians, and Section 5.5 reports how rankings change under four alternatives. A correct answer on an original or random-removal view earns 0.5 more when its box overlaps the lesion with IoU ≥ 0.3 (boxes covering over half the image earn no bonus); unparseable output scores -1; each tool call beyond the first costs 0.05. Section 5.5 reports net utility under four alternative matrices.

3.5 Model and training

The base model is Qwen3.5-4B [34] with LoRA [35] (rank 16, $\alpha = 32$, dropout 0.05, on all linear layers of the language model); the vision encoder is frozen throughout. Training trajectories are generated from the labels alone: optionally a zoom on a jittered box around the lesion (always for lesions under 2% of the image), windowing on 30% of grey-scale images, then the expected final action with short template reasoning. Half of the templated questions are reworded with one of four training paraphrases; three further paraphrases are reserved for testing.

Stage 1, supervised fine-tuning. 16,000 views drawn once at random from the training pools, one

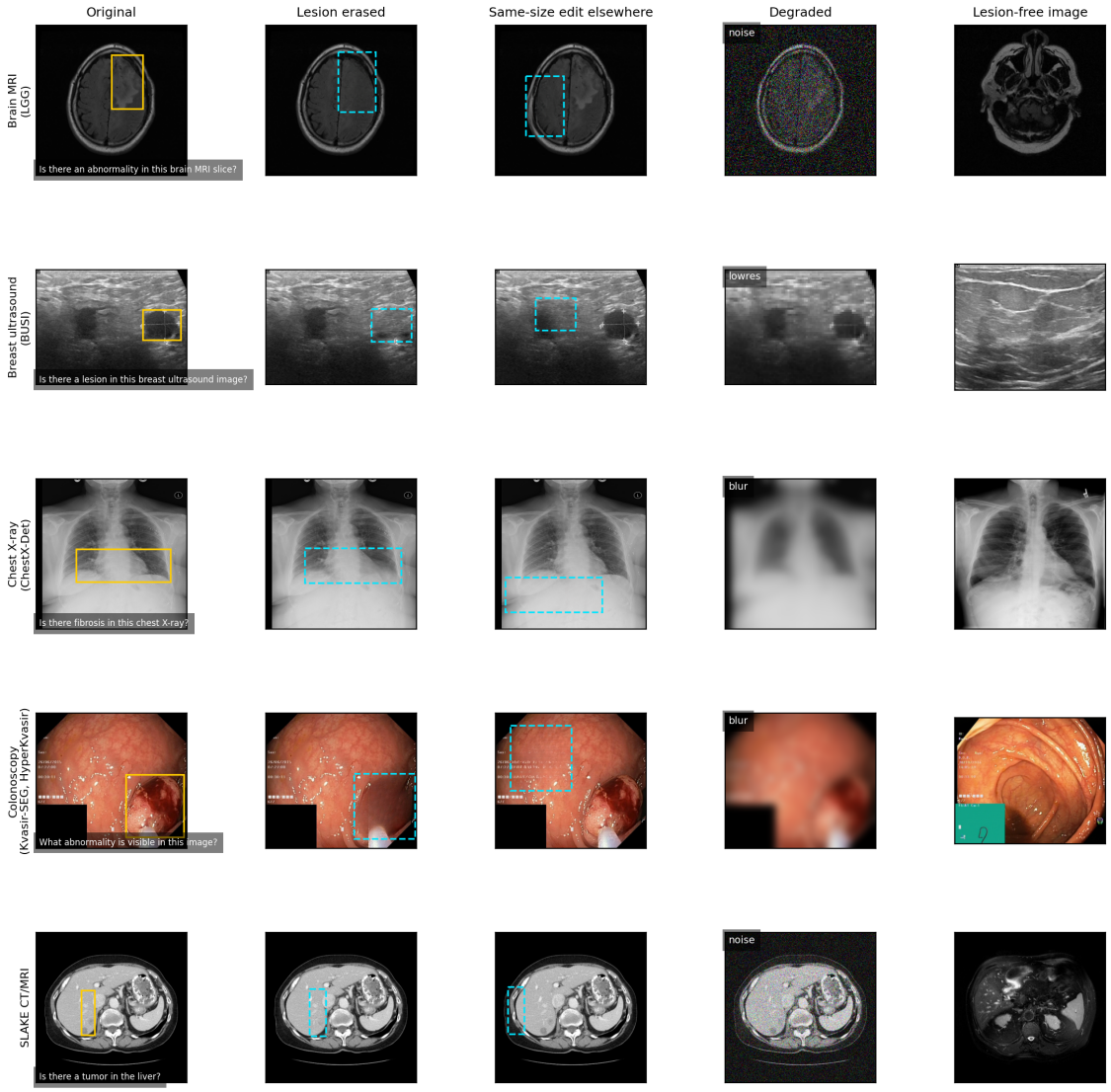


Figure 2: Real test views, one case per imaging type. Yellow: lesion (reference-standard mask); dashed cyan: edited region. Note the second dark region in the breast-ultrasound image that the mask does not cover (Section 8.5).

pass (3,706 optimiser steps), learning rate 2×10^{-4}; cases with lesions under 2% of the image are repeated twice. **Stage 2, rejection-sampling fine-tuning** [28, 29]. The stage-1 model samples $k = 4$ episodes at temperature 1.2 on 200 training views per view kind, 500 lesion-free and 300 lesion-erased views; the best successful episode of each view is kept, and episodes from views the model mostly failed (at most two of four right) are repeated three times. Training continues on these plus 3,000 synthetic views at learning rate 5×10^{-5}. **Stage 3, general-VQA replay.** 5,000 views mixing MedScope views with ordinary questions from the *training* splits of VQA-RAD, SLAKE and PathVQA (any image near-identical to an evaluation image excluded), with "absent" views at no less than 35% of the mix and "Is X healthy?" replay questions balanced between yes and no; learning rate 5×10^{-5}.

3.6 Pre-registered selection

The data plan, recipe, selection rule and reporting rule were committed to the public repository on 1 November 2026, before any v4 data was built or model trained. Let U be validation net utility and H the validation rate of claimed findings on healthy images. Stage 3 is selected if $U_3 \geq U_2 - 0.02$ and $H_3 \leq H_2 + 0.02$;

otherwise stage 2 under the same rule against stage 1; otherwise stage 1. Amendment 1 (after the stage-1 validation result, before any later result) also evaluated the previous release, v3, on the enlarged validation set, to be kept if the chosen stage fell more than 0.02 below it. Amendment 2 (before any test result) added comparisons that cannot change the selection.

Chronology. The main test set and the frontier and open-model runs on it (31 October 2026) predate v4: they were built and run for an earlier release, v3, whose weaknesses motivated the v4 recipe. What was known when the v4 plan was written was therefore the full main-test leaderboard of every comparator and of v3. What was not known was any v4 result; v4 training data, models, validation and test results all postdate the commit. The data and split rules were revised in further commits on the same day, before any v4 data was built (the split check of notebook 22 stopped the first attempt), so the pre-registration is a versioned plan whose history is in the repository, not a single immutable file. The fresh internal and fresh external sets were built as part of v4 and no model, including the comparators, was evaluated on them before the selection. "Tested once" means one locked evaluation of the selected model per test set; the ablation arms and prompting conditions are separate runs on the same cases, reported as such.

On 540 validation cases (1,473 views), U and H were 0.585 and 13.6% for stage 1, 0.610 and 12.7% for stage 2, and 0.613 and 7.7% for stage 3; v3 scored 0.593 and 8.2%. The rule selected stage 3, which is MedScope-4B. It was then evaluated once on every test set; no setting, threshold or prompt was tuned afterwards.

3.7 Pre-registered checks of the mechanism

Two further plans (v5 and v6, in `docs/`) were committed on 6 November 2026, after the main results and before the arms they describe existed, the v6 plan after the v5 results. Each fixes the arms, test sets, endpoints, minimum worthwhile effects and decision rules, and a script applies the rules verbatim.

Unmatched control (C1). The same stage-1 training as the matched-control arm of the factorial, with every control view replaced by an *unmatched* edit: an ellipse of half or twice the erasure's area, random aspect ratio and orientation, placed away from the lesion on tissue, filled with the other training tool (tissue transplant for LaMa erasures, LaMa for transplant erasures); Telea and diffusion stay unseen. Primary endpoint: denial of the intact lesion after a control edit, on the fresh external set and the pooled held-out tools; matching "matters" if the matched arm is lower by at least 5 points with the interval excluding zero on both, within constraints on erased-lesion claims and CCR; the two arms are "within margins" if the difference is under 5 points.

Removal-based preference training (C2). The closest existing method [10], adapted to the same cases and base model: the stage-1 arm trained *without* counterfactual views is continued with direct preference optimisation on pairs built from the recipe's own pools, with no control edits, degraded views or tools: on an erased image the "absent" reply is preferred to the finding with its former box; on an unedited lesion image the reverse; on a lesion-free image "No" to "Yes". All 7,590 usable pairs (the plan named 8,000; open-question lesion-free images have no pair), $\beta = 0.1$, learning rate 2×10^{-5}, batch 8, one pass, with the starting adapter as the reference; the arm therefore receives a full stage 1 *plus* the preference pass, more compute than any SFT arm. Primary endpoints: ESG and net utility on the fresh external set; the recipe "beats" the baseline only if both differences are positive with intervals excluding zero and at least 0.05.

Planned follow-ups (v6). Preference training on top of the recipe (C3: sensitivity up by at least 0.03 without more erased claims), the baseline at two further seeds (C2 over three seeds), and a conditional stage 4 with a validation-only adoption rule; their results are reported when available and cannot change the selected model unless the rule is met.

3.8 Metrics

With O, T, R, N, D the original, target-removal, random-removal, natural-negative and degraded views:

$$\text{ESG} = P(\text{absent answer or refer} \mid T) - P(\text{absent answer or refer} \mid R).$$

CCR (covered-and-correct rate) is the share of answerable views ($O \cup R$) answered correctly; the **strict ESG** counts explicit absent answers only; **erased-lesion claims** is $P(\text{claims finding} \mid T)$ and **healthy-image claims** $P(\text{claims finding} \mid N)$; **grounded accuracy** is correct with IoU ≥ 0.3 on O; the **clarify rate** is $P(\texttt{clarify} \mid D)$; **net utility** is the mean reward over all views. Selective-prediction curves (AURC [30]) are part of the released reports but are not analysed here.

3.9 Statistics

95% confidence intervals use 2,000 bootstrap resamples of cases [36]; a case is one source image with all its views, so the views of a case are resampled together. Model comparisons use a paired bootstrap over shared cases: p is twice the smaller of the two tail proportions of the bootstrap difference distribution, so the smallest reportable value is $1/1000$ and $p < 0.001$ means no resample crossed zero. Seed results are reported as mean $\pm$ standard deviation over three seeds, with each seed pair also compared by paired bootstrap. Proportions on small subsets use Fisher's exact test. The comparisons of ESG, erased-lesion claims and net utility against each of the nine frontier models on the main test are the paper's primary outcomes (27 comparisons). The pre-registration fixed the model, the metrics and the test sets but did not name this family or a multiplicity procedure; the Bonferroni check was added after review, and at its threshold of $0.05/27 = 0.0019$ all 27 remain significant. All other comparisons are secondary and reported unadjusted; those near $p = 0.05$ should be read as exploratory. Where several images come from one patient (BUS-UCLM, BraTS slices, SLAKE volumes) the case bootstrap can understate uncertainty, so every comparison was repeated with whole patients or near-copy groups resampled (Section 5.5).

3.10 Evaluation harness

Open models decode greedily in bf16 on one A100 40 GB; API models use provider defaults. No model receives a model-specific repetition penalty. API models receive PNG images with the long side at most 768 pixels. Answers are parsed from the first final action; matching accepts paraphrases and health-question polarity and reads the asked finding first ("no pneumothorax" is no; "pneumothorax without mediastinal shift" is yes). A reply with no action tag is read as the model's decision, identically for every model; MedScope-4B produced no such reply and no malformed output on any test set, and the per-model rates are in the released reports. Correctness and rewards are recomputed from stored per-view transcripts by one script for every run.

4 Experimental setup

4.1 Study design and data

This is a retrospective study on public, de-identified datasets. The reference standard is each dataset's expert segmentation masks and image labels; expected answers are derived from them by rule (Section 3.2). No human annotation was used for training, selection or scoring.

Training and validation. Training uses SLAKE CT/MRI [7], Kvasir-SEG [37] with HyperKvasir normal landmarks as lesion-free images [38], ChestX-Det [39] (from NIH ChestX-ray14 [40]), BUSI [41] and TCGA-LGG [42]: 1,508 SLAKE and colonoscopy cases plus 1,792 chest X-ray, breast ultrasound and brain MRI cases after quality checks, with 690 lesion-free images from the latter three sources. Validation holds 540 cases (1,473 views). Diffuse chest findings, oversized lesions and brain slices with little tissue are excluded. Table 3 gives the main test set and Table 4 the further test sets, none of which was used for training or selection. Degraded views exist for 170 of the 219 masked cases (the rest failed the view-building checks), which gives 1,115 views.

Protection of the test sets. When the v4 data were built, every existing test image, with its near-copies, was located in its source and checked to be identical (ChestX-Det 93, BUSI 57, TCGA-LGG 79 images, slightly more than the 91, 57 and 78 test cases because near-copies are protected too; none differed or was

Table 3: Main (internal) test set by source.

Source	Imaging	Cases with mask	Lesion-free	Split level
SLAKE [7]	CT / MRI	17	90	near-copy groups
Kvasir-SEG [37] + HyperKvasir [38]	Colonoscopy	83	91	near-copy groups
ChestX-Det [39]	Chest X-ray	41	50	patient
BUSI [41]	Breast ultrasound	42	15	near-copy groups
TCGA-LGG [42]	Brain MRI (FLAIR)	36	42	patient
Total		**219**	**288**	507 cases, 1,115 views

Table 4: Further test sets. "Fresh" sets were untouched until the model was selected (confirmatory).

Test set	Cases	Sources (cases)	Purpose
External centres	169	CVC-ClinicDB [43] (40), QaTa-COV19 [44] (20), BUS-BRA [45] (45), BraTS-Africa 2023 [46, 47] (64, with 50 tumour-free slices)	New scanners, populations
Fresh internal	76	ChestX-Det, unused test-split patients (36 with a lesion, 40 lesion-free)	Confirmatory
Fresh external	316	BraTS 2023 meningioma [48] (82), BUS-UCLM [49] (99), SIIM-ACR pneumothorax [50] (96), ETIS-LaribPolypDB [51] (39)	Confirmatory, new sources
Held-out wordings	373	Main test with three test-only paraphrases	Question wording
Standard VQA	1,051	VQA-RAD test (451), SLAKE English test (300 sampled), PathVQA test (300 sampled)	No regression

missing), and every patient or near-copy group holding one was removed from all v4 splits. Patients seen in earlier training or validation were kept out of the fresh sets, and the fresh sets contain no near-copy of any training or test image. TCGA-LGG contributed no fresh internal cases because all 14 of its test-split patients already hold existing test images. Identifiers do not cross datasets, and some sources share an origin: ChestX-Det (training and main test) and SIIM-ACR (fresh external) both draw on NIH ChestX-ray14, and the BraTS 2020/2021 collections overlap TCGA, which is why the fresh external set uses BraTS 2023 meningioma, a separate cohort with no TCGA-LGG glioma patients. Because SIIM-ACR carries its own patient identifiers rather than NIH ones, same-patient overlap across these datasets is checked at the pixel level: every unedited image of every split and cohort is hashed (64-bit difference hash) and every pair within 8 bits is confirmed by pixel correlation at 64 px, so that the same image, or the same study re-sized or cropped, is found wherever it occurs (`notebooks/31_crosswalk.ipynb`). Among the 6,849 images of the six sets, exactly one image recurs: a validation lesion-free ChestX-Det image (labelled "no finding" there) is the same NIH ChestX-ray14 image as one SIIM-ACR pneumothorax case of the fresh external set (hash distance 0, correlation 1.000). It was never used for training, validation served only for model selection, and as 1 of 316 fresh-external cases it moves no fresh-external mean by more than 0.003; the SIIM result is unaffected because the model answered "No" on every SIIM image (Section 6). The looser tier (within 8 bits, correlation 0.90–0.97) returns 1,834 pairs, but every one lies within the range reached by pairs that cannot share a patient (ChestX-Det images of different patients across splits up to 0.946; SLAKE against BraTS-Africa up to 0.966), so at 64 px this tier reflects shared anatomy rather than shared studies. A different image of the same patient taken on another day is not caught by this check (Section 8.5).

4.2 Models compared

Frontier models through their APIs, identical agent loop and provider defaults, evaluated on 31 October 2026: OpenAI gpt-6-sol, gpt-6-astra, gpt-6.1-sol; Anthropic Claude Fable 5.1, Claude Opus 5.5, Claude Sonnet 5.5; Google gemini-4-argon, gemini-3.1-pro-preview, gemini-3.8-flash. The three flagships (gpt-6-astra, Claude Opus 5.5, gemini-4-argon) were also run on the same 150 main-test cases with a cautious prompt, with four solved training examples in context, at their highest reasoning setting, at their highest image-processing setting, with their own free reply to the plain prompt, and with held-out wordings.

Open models (bf16): Qwen3.5-4B (MedScope's base) [34], MedGemma 1.5 4B [4], Lingshu-7B [5], HuatuoGPT-Vision-7B [2], Qwen2.5-VL-7B [52] and Qwen3-VL-8B [53], each with the agent prompt and, separately, with a generic plain prompt and one free-text reply (Appendix A); model-specific chat templates were applied by each model's processor in both conditions. **Ablation arms**, all at stage 1 with the same number of optimiser steps and the same learning-rate schedule: with counterfactual views (the recipe); without them (originals, degraded and lesion-free views only); without them but with lesion-free images repeated to match the recipe's share of "absent" answers; and a 2×2 factorial of the two training-only components, random-removal control edits and lesion insertions: erasures only (neither), control edits only, insertions only, and both (the recipe). Each arm draws from its own allowed view kinds; the seeds of an arm share that draw and differ in LoRA initialisation, data order and trajectory sampling. The first two arms are trained with three seeds each; both are repeated on MedGemma 1.5 4B. **Trivial policies** that ignore the image: always claim the finding, always say it is absent, always refer, always clarify.

4.3 Compute and software

Every training stage ran on a single NVIDIA A100 40 GB (Google Colab), about 4 hours per stage; one evaluation of a local model on the main test takes about 1 hour. Software: PyTorch 2.11, Transformers 5.22.1, PEFT 0.21.2, bitsandbytes 0.50.2. The nine frontier runs on the main test cost US$132 in API fees at list prices of 31 October 2026 [total over all frontier runs, from the stored token counts].

5 Results

5.1 Main test

MedScope-4B has the highest ESG (0.758 [0.701, 0.815]) of all 16 models (Table 5). It is better than all nine frontier models on ESG, erased-lesion claims and net utility, and than all six open models on ESG and net utility (all $p < 0.001$; Table 6). Its erased-lesion claims (9.6% [6.1, 13.7]) are 32–52 points below every frontier model's; one open model claims fewer (Qwen2.5-VL-7B 5.5%, $p = 0.11$) and one not significantly more (Qwen3-VL-8B 12.8%, $p = 0.33$), because both rarely answer lesion questions correctly at all (CCR 0.14 and 0.55), which is why ESG, not the claim rate alone, is the primary metric. Accuracy on real lesions does not differ significantly from any frontier model (CCR 0.849 [0.804, 0.894] vs 0.829–0.888; all $p \geq 0.12$); the confidence intervals of the differences (Table 6) reach -0.09, so equivalence is not established. On unedited healthy images it reports a finding in 7.6% [4.8, 10.9] of cases, fewer than eight of nine frontier models ($p \leq 0.001$); against gemini-4-argon (9.0%) the difference is not significant ($p = 0.59$). It asks to clarify on every degraded view (170 of 170).

5.2 By imaging type

The ESG and erased-lesion advantages over the three flagships hold in every imaging type (Table 7); net utility does not, since on SLAKE CT/MRI MedScope-4B (0.22) is below gemini-4-argon (0.38) and Claude Opus 5.5 (0.32). MedScope-4B is strongest on colonoscopy (ESG 0.95), brain MRI and breast ultrasound (0.74–0.75), and weakest on chest X-ray (0.51) and SLAKE CT/MRI (0.47). No flagship exceeds ESG 0.69 in any type, and on chest X-ray the three flagships lie between -0.05 and 0.20.

5.3 What produces the behaviour

The counterfactual views are the active ingredient. Removing them from stage-1 training, with the same data draw, optimiser steps and schedule, lowers ESG on every seed (seed by seed $+0.525$, $+0.539$ and $+0.324$ in favour of the recipe, all $p < 0.001$) and raises erased-lesion claims to the frontier level or above (Table 10). The arm without counterfactual views says "present" more readily, so it is more accurate on

Table 5: Main test (507 cases, 1,115 views; agent prompt for every model). Erased, Healthy: % of views on which the model claims the finding after the lesion was erased (219 views), or on an unedited lesion-free image (288 views); CCR over the 438 original and random-removal views. 95% bootstrap CI for MedScope-4B's net utility; its other CIs are in the text.

Model	CCR	ESG	Erased %	Healthy %	Net utility
MedScope-4B (this work)	0.849	**0.758**	**9.6**	**7.6**	**0.732** [0.649, 0.813]
gemini-4-argon	0.874	0.470	41.1	9.0	0.565
Claude Sonnet 5.5	0.870	0.402	48.4	16.7	0.381
gemini-3.1-pro-preview	0.849	0.384	51.6	26.7	0.306
gpt-6-sol	0.829	0.342	51.6	25.7	0.261
Claude Fable 5.1	0.856	0.342	52.1	23.3	0.255
Claude Opus 5.5	0.863	0.329	53.9	23.6	0.358
gpt-6.1-sol	0.888	0.324	58.9	30.9	0.276
gpt-6-astra	0.886	0.306	58.9	27.8	0.309
gemini-3.8-flash	0.868	0.269	61.6	27.8	0.263
Lingshu-7B	0.527	0.416	32.0	12.5	−0.249
Qwen3.5-4B (base)	0.758	0.338	42.0	19.4	0.134
Qwen3-VL-8B	0.553	0.297	12.8	12.5	0.186
MedGemma 1.5 4B	0.674	0.283	54.3	24.0	−0.255
HuatuoGPT-Vision-7B	0.498	0.265	30.1	22.9	−0.426
Qwen2.5-VL-7B	0.144	0.146	5.5	24.7	−0.742
Always claims the finding	1.000	0.000	100	100	−0.478
Always says it is absent	0.000	0.000	0	0	−0.848
Always refers	0.000	0.000	0	0	0.110

Table 6: MedScope-4B minus each model, paired bootstrap over the same 507 cases: difference [95% CI]. Positive is better for ESG and net utility; negative is better for the claim rates. Against the other four open models every ESG and net-utility difference is also $p < 0.001$.

Compared with	ESG	Erased-lesion claims	Healthy-image claims	CCR	Net utility
gemini-4-argon	+0.288 [0.208, 0.367]	−0.315 [−0.385, −0.243]	−0.014 [−0.059, 0.028]	−0.025 [−0.070, 0.017]	+0.168 [0.083, 0.254]
Claude Sonnet 5.5	+0.356 [0.284, 0.431]	−0.388 [−0.459, −0.318]	−0.090 [−0.137, −0.045]	−0.021 [−0.069, 0.026]	+0.351 [0.267, 0.437]
gemini-3.1-pro-prev.	+0.374 [0.298, 0.448]	−0.420 [−0.491, −0.350]	−0.191 [−0.252, −0.133]	0.000 [−0.043, 0.044]	+0.426 [0.337, 0.519]
gpt-6-sol	+0.416 [0.341, 0.489]	−0.420 [−0.491, −0.346]	−0.181 [−0.235, −0.126]	+0.021 [−0.022, 0.063]	+0.472 [0.389, 0.554]
Claude Fable 5.1	+0.416 [0.339, 0.496]	−0.425 [−0.500, −0.355]	−0.156 [−0.212, −0.102]	−0.007 [−0.054, 0.039]	+0.477 [0.384, 0.566]
Claude Opus 5.5	+0.429 [0.345, 0.507]	−0.443 [−0.515, −0.368]	−0.160 [−0.217, −0.103]	−0.014 [−0.066, 0.036]	+0.374 [0.283, 0.466]
gpt-6.1-sol	+0.434 [0.354, 0.520]	−0.493 [−0.566, −0.423]	−0.233 [−0.292, −0.178]	−0.039 [−0.088, 0.010]	+0.456 [0.360, 0.552]
gpt-6-astra	+0.452 [0.364, 0.539]	−0.493 [−0.562, −0.426]	−0.201 [−0.261, −0.145]	−0.037 [−0.087, 0.014]	+0.424 [0.329, 0.522]
gemini-3.8-flash	+0.489 [0.407, 0.566]	−0.521 [−0.596, −0.446]	−0.201 [−0.259, −0.143]	−0.018 [−0.065, 0.029]	+0.470 [0.380, 0.560]
Qwen3.5-4B (base)	+0.420 [0.338, 0.498]	−0.324 [−0.398, −0.249]	−0.118 [−0.171, −0.070]	+0.091 [0.035, 0.148]	+0.599 [0.503, 0.695]
MedGemma 1.5 4B	+0.475 [0.391, 0.556]	−0.447 [−0.518, −0.376]	−0.163 [−0.223, −0.104]	+0.176 [0.116, 0.235]	+0.988 [0.878, 1.099]

Table 7: Main test by imaging type: ESG / erased-lesion claims (fraction) / net utility.

Imaging (cases)	MedScope-4B	gemini-4-argon	Claude Opus 5.5	gpt-6-astra
Colonoscopy (174)	**0.95** / 0.01 / 1.06	0.69 / 0.31 / 0.83	0.57 / 0.40 / 0.53	0.40 / 0.58 / 0.40
Brain MRI (78)	**0.75** / 0.14 / 0.75	0.42 / 0.47 / 0.56	0.31 / 0.61 / 0.51	0.50 / 0.31 / 0.43
Breast ultrasound (57)	**0.74** / 0.14 / 0.78	0.40 / 0.48 / 0.53	0.29 / 0.62 / 0.40	0.12 / 0.83 / 0.28
Chest X-ray (91)	**0.51** / 0.17 / 0.40	0.20 / 0.49 / 0.21	−0.05 / 0.66 / −0.12	0.15 / 0.66 / 0.14
SLAKE CT/MRI (107)	**0.47** / 0.12 / 0.22	0.35 / 0.41 / 0.38	0.24 / 0.59 / 0.32	0.29 / 0.47 / 0.17

real lesions (CCR 0.955 vs 0.871) and fails whenever the lesion is gone. Healthy-image claims also fall with the recipe (13.9% vs 35.9%), although both arms train on the same lesion-free images.

It is not the share of "No" answers. Repeating lesion-free images until the share of "absent" targets equals the recipe's leaves ESG at 0.215 (recipe 0.735; difference +0.521 [0.440, 0.600], $p < 0.001$).

The control edit is the component that matters; the evidence on insertions is mixed. In the 2 × 2 factorial (Table 8; single seed) ESG lies between 0.71 and 0.74 in all four arms and no pair differs

significantly, so the erasures account for the evidence sensitivity as far as one seed can show. The control edit raises accuracy on real lesions (CCR $+0.109$ without insertions and $+0.041$ with, $p \leq 0.018$) at the price of more erased-lesion claims ($+8$ and $+5$ points), and it is what reduces denial of an intact lesion after an edit elsewhere, the "edited means absent" shortcut: on an unseen tool (Telea), the two arms without control edits deny the intact lesion after an edit elsewhere in 43% of cases, the two with it in 16% and 23% ($p < 0.001$). Insertions have no measurable benefit on the main test (CCR $+0.018$ without control edits and -0.050 with; ESG unchanged) or on the held-out tools, but on the fresh external set the full recipe has higher net utility than control edits alone ($+0.144$ [0.048, 0.240], $p = 0.002$), through fewer healthy-image claims (6.8% vs 13.5%). Their effect therefore depends on the endpoint and interacts with the control edit; the recipe is kept as registered and this is reported.

Pre-registered check C1: matching the control edit is not shown to matter. Table 9 gives the matched and unmatched arms side by side. On the main test, the pooled held-out tools and three of the four fresh-external sources the two are within seed noise on every endpoint (denial 10.5% vs 9.1% on the main test, 13.7% vs 12.9% on the held-out tools; ESG 0.744 vs 0.753 and 0.749 vs 0.761). The only difference is SIIM pneumothorax, where the unmatched arm answers "present" on 60% of unedited films but also on 67% of erased and 19% of healthy ones, a response bias on a finding it does not see (its SIIM ESG is about zero), while the matched arm answers "No" on all of them; because the cost matrix charges a missed critical finding more than a false one, this bias raises the unmatched arm's fresh-external net utility by 0.25. The pre-registered rule returns "inconclusive" (the primary endpoint moved, in the opposite direction, for this reason); the substantive reading is that the benefit belongs to training with control edits, not to their matching, and the paper's claims are narrowed accordingly.

Pre-registered check C2: the recipe is not shown to beat removal-based preference training; the two fail differently. On the fresh external set the DPO baseline reaches the recipe's ESG (0.369 vs 0.381, $+0.012$ [-0.053, 0.077]), so by the rule the recipe is not shown to be better on the paper's primary metric on new data. The recipe has higher net utility there ($+0.148$ [0.054, 0.247], $p < 0.001$), through higher sensitivity on BraTS 2023 meningioma (0.625 vs 0.438; both arms answer "No" on every SIIM film) and fewer denials, and on the main test it has higher ESG ($+0.078$, $p = 0.012$) and net utility ($+0.090$, $p = 0.022$) with half the erased-lesion claims (11.9% vs 23.7%). The baseline is more sensitive to real lesions (CCR 0.900 vs 0.847 on the main test, 0.895 vs 0.810 on the held-out tools) and denies fewer intact lesions after an edit elsewhere (10.0% vs 14.6%), although it never saw a control edit. Preference training and counterfactual supervised training therefore repair different errors; whether they combine is the pre-registered C3 (Section 3.7).

It replicates on a second base model. The same recipe on MedGemma 1.5 4B gives ESG 0.721 vs 0.374 without counterfactual views ($+0.347$ [0.276, 0.425], $p < 0.001$) and erased-lesion claims 7.8% vs 57.5%. With counterfactual views the two base models do not differ (net utility -0.005, $p = 0.83$; ESG $+0.014$, $p = 0.73$).

Stages 2 and 3 add little on the main test. The final model against its own stage 1: net utility $+0.040$ ($p = 0.15$), ESG $+0.023$ ($p = 0.39$), healthy-image claims -1.7 points ($p = 0.35$). Their contribution is on validation healthy-image claims, which the selection rule used (13.6% $\rightarrow$ 7.7%), and on ordinary VQA questions (Section 5.6).

The recipe's effect also holds at the external centres (three seeds: ESG 0.622 ± 0.030 vs 0.389 ± 0.043; erased-lesion claims 14.6% vs 48.2%; healthy-image claims 9.3% vs 49.3%) and on the fresh external test (net utility 0.070 ± 0.008 vs -0.149 ± 0.033; healthy-image claims 9.7% vs 26.1%), where the ESG difference is smaller (0.381 vs 0.288) and significant on two of three seeds.

5.4 Does it read the lesion or the edit?

Every training erasure used LaMa or tissue transplant. If MedScope-4B had learned those tools' traces, its advantage should shrink on other tools; persistence tests transfer beyond the training tools but cannot rule out cues the tools share. We remade both edited views of 125 main-test cases on the same masks with Telea [19] and Stable Diffusion inpainting [21], neither used in training or selection, and with tissue

Table 8: 2×2 factorial of the training-only components, stage 1, seed 0: main test (507 cases) and denial of the intact lesion after a Telea edit elsewhere (125 cases; Telea was never used in training).

Control edits	Insertions	CCR	ESG	Erased %	Healthy %	Denies after Telea edit %
no	no (erasures only)	0.788	0.708	6.8	5.9	43.2
no	yes	0.806	0.735	7.3	11.1	43.2
yes	no	**0.897**	**0.744**	15.1	11.8	**16.0**
yes	yes (the recipe)	0.847	0.735	11.9	9.4	23.2

Table 9: Pre-registered checks C1 and C2 (stage 1, seed 0). Denial: intact lesion denied or referred after a control edit. Held-out tools: Telea and diffusion edits of the main test, pooled (219 cases). Fresh external: 316 cases, of which 96 SIIM pneumothorax films on which every arm except the unmatched one answers "No".

Set	Arm	CCR	ESG	Erased %	Denial %	Net utility
Main test	Matched control (factorial arm)	0.897	0.744	15.1	10.5	0.727
	Unmatched control	0.904	0.753	15.1	9.1	0.739
	Recipe	0.847	0.735	11.9	14.6	0.692
	No CF, then removal-based DPO	0.900	0.658	23.7	10.0	0.602
Held-out tools	Matched control	0.859	0.749	11.0	13.7	0.733
	Unmatched control	0.868	0.761	10.5	12.9	0.772
	Recipe	0.810	0.649	16.3	18.8	0.566
	No CF, then removal-based DPO	0.895	0.666	23.4	10.0	0.644
Fresh external	Matched control	0.488	0.405	8.3	51.2	-0.083
	Unmatched control	0.670	0.411	27.4	31.5	0.165
	Recipe	0.560	0.381	16.7	45.2	0.061
	No CF, then removal-based DPO	0.461	0.369	8.9	54.2	-0.087

Table 10: Stage-1 arms on the main test (507 cases). Three-seed arms: mean $\pm$ sd.

Arm	CCR	ESG	Erased %	Healthy %	Net utility
Qwen3.5-4B, with counterfactual views (3 seeds)	0.871 ± 0.022	$\mathbf{0.741 \pm 0.019}$	13.2 ± 3.2	13.9 ± 4.0	$\mathbf{0.684 \pm 0.021}$
Qwen3.5-4B, without (3 seeds)	0.955 ± 0.008	0.279 ± 0.107	67.6 ± 11.9	35.9 ± 3.9	0.341 ± 0.090
Without, "absent" share matched (seed 0)	0.945	0.215	73.5	31.6	0.321
Erasures only (seed 0)	0.788	0.708	6.8	5.9	0.667
Control edits only, matched (seed 0)	0.897	0.744	15.1	11.8	0.727
Control edits only, unmatched (seed 0)	0.904	0.753	15.1	12.5	0.739
Without, then removal-based DPO (seed 0)	0.900	0.658	23.7	24.3	0.602
MedGemma 1.5 4B, with counterfactual views	0.806	0.721	7.8	5.9	0.698
MedGemma 1.5 4B, without	0.947	0.374	57.5	31.6	0.409
MedGemma 1.5 4B, untrained	0.674	0.283	54.3	24.0	-0.255

transplant (Table 11).

MedScope-4B beats all three flagships on every tool (ESG differences $+0.10$ to $+0.48$, $p \leq 0.049$). On Telea its erased-lesion claims are unchanged (7.2%; ESG -0.088, $p = 0.025$, from more denials after the edit elsewhere); on Stable Diffusion they rise by 10.4 points ($p = 0.006$), the largest drop, yet still two to four times below Claude Opus 5.5 and gpt-6-astra. The arms without counterfactual views stay near ESG 0.1–0.3 on every tool. gemini-4-argon's erased-lesion claims fall from 33% to 9% on Telea only, whose smooth fills it recognises as edits: its absent answers partly follow visible traces.

Clean-looking erasures. Asked for the probability that an image was edited, gemini-4-argon's ratings of the 219 erased views are bimodal (61 rated ≤ 0.20, 158 rated ≥ 0.75, none in between), so the split needs no threshold choice. On erasures it rates as unedited, MedScope-4B still claims the lesion in only 15% (8% on visible ones), against 41–64% for the nine frontier models, which claim the lesion as often on visible erasures (41–58%).

Edit visibility. A ResNet-18 trained on 5,126 training views separates random-removal from original

Table 11: Edited views by inpainting tool, 125 cases: ESG / erased-lesion claims (%).

Model	LaMa (training)	Transplant (training)	Telea (unseen)	Stable Diffusion (unseen)
MedScope-4B	**0.752** / 7.2	0.720 / 8.0	0.664 / 7.2	0.640 / 17.6
gemini-4-argon	0.536 / 32.8	0.488 / 32.8	0.560 / 8.8	0.392 / 40.0
Claude Opus 5.5	0.376 / 45.6	0.264 / 56.8	0.296 / 41.6	0.352 / 46.4
gpt-6-astra	0.312 / 55.2	0.240 / 64.0	0.256 / 60.8	0.256 / 61.6
Stage 1, the recipe	0.720 / 12.0	**0.744** / 8.0	0.696 / 7.2	0.568 / 24.8
Stage 1, control edits only	0.712 / 17.6	0.736 / 9.6	**0.768** / 7.2	**0.704** / 16.0
Stage 1, insertions only	0.696 / 6.4	0.664 / 4.8	0.512 / 5.6	0.600 / 11.2
Stage 1, erasures only	0.688 / 7.2	0.656 / 4.8	0.544 / 2.4	0.712 / 4.8
Stage 1, without counterfactual views	0.272 / 69.6	0.200 / 76.8	0.144 / 83.2	0.232 / 73.6
Stage 1, without, "absent" share matched	0.224 / 72.0	0.208 / 73.6	0.096 / 85.6	0.184 / 77.6

views with AUC 0.83 [0.80, 0.87], so the inpainting leaves a trace that a trained detector can find. A linear probe on the frozen vision encoder's own output, the features the adapter receives, finds far less of it: random-removal vs original AUC 0.60 [0.58, 0.63] (logistic regression on mean-pooled features, 1,200 training views, 438 test views), while target-removal vs random-removal views, which differ in the lesion and in any anatomy-dependent trace of its erasure, separate at AUC 0.86 [0.83, 0.89]. This measures the information recovered by one regularised linear probe on pooled features; a differently fitted or nonlinear reader could recover more. Zero-shot, gemini-4-argon and gpt-6.1-sol cannot separate random-removal views from originals (AUC 0.52–0.55) but find erased-lesion views suspicious (0.70–0.75): erasing a lesion leaves a smooth region where structure is expected.

Four results argue that MedScope's answers follow the lesion more than edit traces: (i) a policy that responds to the shared editing signature alike in both views scores ESG 0, and MedScope-4B denies the lesion on only 14.6% of random-removal views, which carry the same tool traces; (ii) the vision encoder is frozen, so the adapter cannot learn new pixel-level features of the kind the ResNet-18 learns, and a linear probe finds only weak edit information in the frozen features (AUC 0.60); (iii) the advantage holds on clean-looking erasures and on two unseen tools; (iv) on completely unedited lesion-free images, where no edit hypothesis applies, the recipe lowers false findings from 35.9% to 13.9% (Table 10). None of these excludes a residual contribution of anatomy-dependent erasure traces, which the frontier models' higher suspicion of erased than of control views shows to exist.

5.5 Robustness

Lesion size. Evidence sensitivity rises with lesion area, and the smallest lesions remain the weakest (Figure 3; erased-lesion claims 18.2% in the smallest quartile and 1.8% in the largest). The flagships show no such trend: their ESG stays between 0.20 and 0.56 in every quartile, and gpt-6-astra claims a large erased lesion in 75% of cases.

Prompting the frontier models harder. MedScope learned from thousands of counterfactual views; the frontier models saw none. On the same 150 cases we gave the three flagships a cautious prompt (an explicit warning that the question may name an absent finding), four solved training examples in context (a lesion; the same image erased; an edit elsewhere with the lesion intact; a healthy image), and their highest reasoning setting. None closes the gap (Table 12). At the highest reasoning setting no flagship differs from its default (all $p \geq 0.13$). MedScope-4B stays better than every flagship under every condition on ESG ($p \leq 0.001$) and net utility ($p \leq 0.034$).

Image-processing setting and prompt format. Every model receives the same PNG (long side at most 768 pixels, the size at which the benchmark images are stored), but OpenAI and Google also choose how finely an image is processed. At their highest setting (OpenAI image detail "high", Gemini media resolution "high"; Anthropic has no such setting, as images up to 1,568 pixels are processed as sent) neither gpt-6-astra nor gemini-4-argon changes measurably: ESG 0.338 $\rightarrow$ 0.397 ($p = 0.18$) and 0.441 $\rightarrow$ 0.485 ($p = 0.34$), net utility within 0.02 of the default. With the generic plain prompt (one free reply, no action format, no

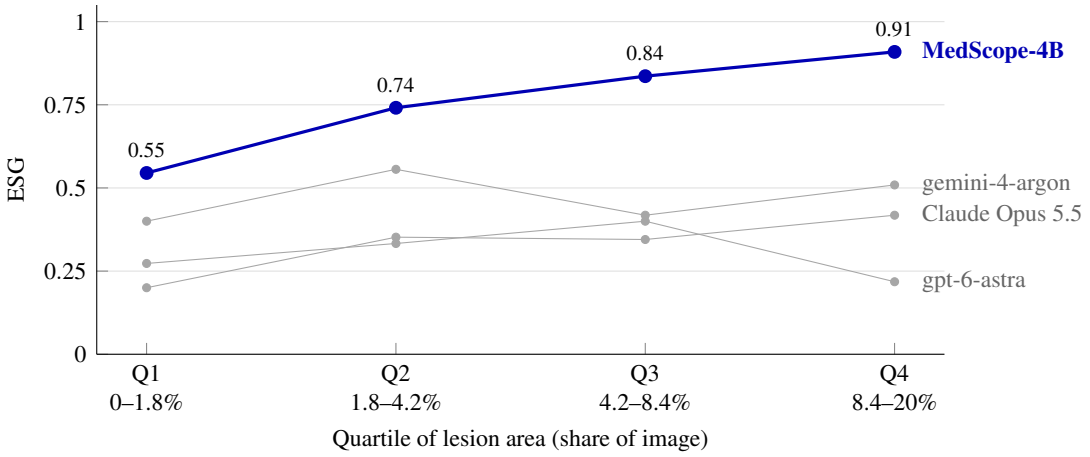


Figure 3: Evidence sensitivity gap by quartile of lesion area on the main test (55, 54, 55, 55 cases with a lesion mask). The gap between MedScope-4B and the flagships grows with lesion size but holds in the smallest quartile (0.55 vs 0.20–0.40).

Table 12: Flagships under six conditions, same 150 cases: ESG / erased-lesion claims (%). MedScope-4B on these cases: 0.750 / 7.4. Highest image setting: OpenAI image detail "high", Gemini media resolution "high"; Anthropic has no such setting. Plain prompt: one free reply, no tools.

Model	Default	Cautious	Solved examples	Highest reasoning	Highest image setting	Plain prompt
gemini-4-argon	0.441 / 41.2	0.412 / 42.6	0.456 / 36.8	0.456 / 39.7	0.485 / 38.2	0.338 / 52.9
Claude Opus 5.5	0.338 / 54.4	0.382 / 44.1	0.309 / 52.9	0.397 / 51.5	–	0.412 / 38.2
gpt-6-astra	0.338 / 52.9	0.397 / 47.1	0.500 / 38.2	0.324 / 58.8	0.397 / 51.5	0.250 / 66.2

tools) no flagship gains net utility: gpt-6-astra falls from 0.33 to -0.05 and gemini-4-argon from 0.55 to 0.28 (both $p < 0.001$), Claude Opus 5.5 from 0.39 to 0.26 ($p = 0.061$). The effect on erased-lesion claims differs by model: gpt-6-astra claims 66% of erased lesions instead of 53% ($p = 0.023$) and 55% of findings on healthy images instead of 33% ($p < 0.001$), gemini-4-argon 53% instead of 41% ($p = 0.083$), whereas Claude Opus 5.5 claims fewer (38% instead of 54%, $p = 0.002$; healthy images 17% instead of 26%, $p = 0.011$) at the cost of fewer correct answers (CCR 0.85 vs 0.89). Under the plain prompt MedScope-4B leads every flagship by 0.34–0.50 in ESG and 0.45–0.78 in net utility ($p < 0.001$). The agent loop therefore does not handicap the flagships on the outcome that combines both error types.

Open models with a plain prompt. With the generic plain prompt and one free-text reply, every open model claims erased lesions more often than under the agent prompt (Qwen3.5-4B 72%, MedGemma 1.5 4B 72%, HuatuoGPT-Vision-7B 60%, Qwen2.5-VL-7B 56%, Qwen3-VL-8B 56%, Lingshu-7B 53%; these runs do not involve MedScope and are unchanged since the v3 evaluation). The same holds for the flagships (previous paragraph). This tests one alternative prompt, not each model's full native protocol, so it argues against, without excluding, a handicap from the agent loop.

Question wording. On 373 cases with three test-only paraphrases, net utility is unchanged against the original wording ($+0.006$ [-0.033, 0.041], $p = 0.74$). With the same paraphrases on 150 cases, MedScope-4B leads the flagships by $+0.35$ to $+0.57$ in net utility ($p < 0.001$).

Tools off. With every tool call refused at test time, ESG is unchanged (0.753 vs 0.758, $p = 0.92$) and net utility falls by 0.092 ($p = 0.003$): the model says "absent" more often on lesion views (21% vs 15%). Evidence sensitivity comes from training; the tools add accuracy on real lesions. MedScope-4B calls a tool on 0.99 of lesion views (35% of calls are windowing), and its first zoom covers the lesion in 93%. gpt-6-sol and gpt-6-astra almost never call a tool.

Cost matrix. The always-refer policy earns $+0.110$ under the published matrix, a weakness of that matrix; with referral on lesion-free images paid 0 it earns -0.118, and -0.300 when referral costs -0.3 everywhere. MedScope-4B ranks first among all models under all five matrices (wrong-answer penalty -2,

Table 13: MedScope-4B and the three flagships on the further test sets.

Test set	Model	CCR	ESG	Erased %	Healthy %	Net utility
External centres (169)	**MedScope-4B**	0.727	**0.597**	**11.8**	**4.0**	0.530
	gemini-4-argon	0.950	0.370	58.8	18.0	0.544
	Claude Opus 5.5	0.966	0.412	53.8	30.0	0.544
	gpt-6-astra	0.945	0.277	66.4	24.0	0.424
Fresh internal (76)	**MedScope-4B**	0.681	**0.472**	**19.4**	**12.5**	**0.402**
	gemini-4-argon	0.639	0.056	61.1	12.5	0.048
	Claude Opus 5.5	0.653	0.000	63.9	25.0	−0.023
	gpt-6-astra	0.847	0.083	75.0	17.5	0.219
Fresh external (316)	**MedScope-4B**	0.476	0.393	**8.3**	**4.7**	−0.053
	gemini-4-argon	0.741	0.435	31.5	9.5	**0.310**
	Claude Opus 5.5	0.696	0.292	42.3	18.9	0.056
	gpt-6-astra	0.634	0.268	35.7	34.5	−0.106

-1, -0.5; the two referral variants), and its strict ESG equals its ESG (0.758), since it never refers.

Degraded views. Without the degraded views, which favour a model trained to clarify, net utility is 0.774 for MedScope-4B and 0.310–0.579 for the flagships (MedScope better, $p < 0.001$).

Label noise. Without human review, the nine frontier models served as a comparator-model consensus panel, independent of MedScope but not a clinical reference standard: a lesion-free test image that at least seven call abnormal is treated as a disputed label (26 of 288: 12 colonoscopy, 11 SLAKE, 2 breast ultrasound, 1 chest X-ray; MedScope does not vote). Without them, MedScope-4B's healthy-image claims are 6.9% (gemini-4-argon 4.6%, other frontier models 10.3–24.4%). Every comparison in Table 6 keeps its direction and significance except one: the healthy-image difference against gemini-4-argon changes sign (-0.014, $p = 0.59$ before; $+0.023$, $p = 0.31$ after) and stays non-significant.

Dependence between cases. The 507 main-test cases come from 416 patients or near-copy groups (39 groups with more than one case, the largest with 8); the external centres from 155 (169 cases), the fresh internal set from 75 (76) and the fresh external set from 230 (316; largest group 8). Resampling whole groups instead of cases widens the 95% intervals by at most 0.01 on the main test and 0.03 on the fresh external set. All 27 primary comparisons survive the Bonferroni threshold with $p \leq 0.001$ (26 with no resample crossing zero; net utility against gemini-4-argon at $p = 0.001$), and of every comparison in the paper exactly one crosses $p = 0.05$: fresh-external ESG against Claude Opus 5.5 ($p = 0.037$ by cases, 0.060 by patients).

5.6 Generalisation

On every new test set MedScope-4B keeps the lowest erased-lesion claims and the lowest (once tied) healthy-image claims of the four models (Table 13). The flagships detect more real lesions at the external centres and on the fresh external set; on the fresh internal set the CCR comparison is mixed (0.68 vs 0.64–0.85). Net utility, which weighs both, does not differ at the external centres, favours MedScope-4B on the fresh internal set, and favours gemini-4-argon on the fresh external set.

External centres. ESG is higher than every flagship's ($p \leq 0.002$), erased-lesion claims are 42–55 points lower ($p < 0.001$) and healthy-image claims 14–26 points lower ($p \leq 0.026$), while CCR is 0.22–0.24 lower ($p < 0.001$); net utility does not differ ($p \geq 0.07$). By source, ESG is 0.85 on CVC-ClinicDB, 0.71 on BraTS-Africa, 0.58 on BUS-BRA and 0.05 on QaTa-COV19, where diffuse COVID-19 opacities are erased far less completely than focal lesions.

Fresh internal (confirmatory, chest X-ray). ESG 0.47 against 0.00–0.08 for the flagships ($p \leq 0.001$); net utility better than Claude Opus 5.5 and gemini-4-argon ($p = 0.002$), not different from gpt-6-astra ($p = 0.15$).

Fresh external (confirmatory). The result is mixed. ESG is higher than gpt-6-astra's ($p = 0.005$),

Table 14: Fresh external test by source: CCR / net utility.

Source (cases)	MedScope-4B	gemini-4-argon	Claude Opus 5.5	gpt-6-astra
BUS-UCLM, breast ultrasound (99)	0.73 / **0.67**	0.71 / 0.55	0.93 / 0.48	0.98 / 0.18
ETIS-Larib, colonoscopy (39)	0.69 / 0.38	0.97 / **0.75**	0.92 / 0.59	0.73 / 0.35
BraTS 2023 meningioma, MRI (82)	0.53 / −0.03	0.62 / **0.09**	0.69 / 0.00	0.50 / −0.21
SIIM-ACR pneumothorax, chest X-ray (96)	0.00 / −1.10	0.66 / **−0.06**	0.28 / −0.68	0.29 / −0.62

Table 15: Standard medical VQA, same agent loop for every model: closed / open / all questions.

Model	VQA-RAD (451)	SLAKE (300)	PathVQA (300)
MedScope-4B	0.741 / 0.230 / 0.514	0.881 / 0.644 / 0.730	0.801 / 0.125 / 0.477
gemini-4-argon	0.884 / 0.410 / 0.674	0.927 / 0.696 / 0.780	0.712 / 0.097 / 0.417
Claude Opus 5.5	0.876 / 0.380 / 0.656	0.927 / 0.602 / 0.720	0.865 / 0.153 / 0.523
gpt-6-astra	0.797 / 0.255 / 0.557	0.835 / 0.476 / 0.607	0.526 / 0.062 / 0.303
Qwen3.5-4B (base)	0.657 / 0.260 / 0.481	0.826 / 0.424 / 0.570	0.545 / 0.021 / 0.293
Lingshu-7B	0.705 / 0.195 / 0.479	0.798 / 0.545 / 0.637	0.712 / 0.042 / 0.390

Table 16: SIIM pneumothorax images called present (the 42 cases found in the original-resolution data).

Model	256-pixel copy	Original resolution
Qwen3.5-4B (base)	0%	12%
Stage 1 without counterfactual views	0%	5%
Stage 1 with counterfactual views	19%	40%
Stage 2	[–]	40%
MedScope-4B (stage 3)	**0%**	**0%**
gemini-4-argon	69%	69%

marginally higher than Claude Opus 5.5's ($p = 0.037$ by cases, 0.060 by patients) and not different from gemini-4-argon's ($p = 0.34$); net utility is below gemini-4-argon's (-0.363 [-0.496, -0.233], $p < 0.001$). The by-source breakdown (Table 14) locates the loss: MedScope-4B leads on breast ultrasound and trails on colonoscopy and meningioma MRI, and on SIIM pneumothorax it answers "No" to every one of 48 pneumothorax images (Section 6).

Standard medical VQA. Accuracy on ordinary questions is kept (Table 15). MedScope-4B saw these benchmarks' training splits during replay, so this is not a like-for-like comparison with the zero-shot models; it shows that aggregate accuracy on these benchmarks exceeds the base model's on all three after the evidence training, not that every question type is preserved. With their own plain prompt most open models score higher than in the agent loop (Lingshu-7B: 0.552, 0.803, 0.573), so the loop is a harder format for open models on ordinary VQA.

6 Failure analysis: why is pneumothorax missed?

MedScope-4B answered a confident "No" on all 48 SIIM-ACR pneumothorax images of the fresh external test (answer-token probability about 1.0). Scoring and parsing were checked and are correct. The analysis below was made after the test results, is labelled post hoc in the pre-registration, and changes no model or confirmatory number. It trains nothing: every number comes from stored runs or from rebuilding each stage's exact training list with the training code.

Resolution does not explain it. The fresh test used a 256-pixel copy of SIIM. Rebuilt from the original DICOMs, earlier checkpoints detect more pneumothoraxes, but the final model still detects none of the 42 recovered cases (Table 16; exact 95% CI for 0/42: 0–8.4%).

Counterfactual training does not explain it. The stage-1 model with counterfactual views is the most sensitive Qwen-based model at both resolutions; the arm without them detects almost none. An interaction

with the later stages is not excluded.

Training coverage is the most plausible explanation. Questions naming pneumothorax were 54 of about 7,000 chest X-ray training views in stage 1 (35 in stage 2, 15 in stage 3), trained to "No" 80%, 100% and 87% of the time. Stage 1 held only four pneumothorax lesion images, all large (bounding box median 37% of the image, against 6.9% for the SIIM cases; the 20% cap of Section 3.2 applies to mask area, not box area). No validation or internal test set held a pneumothorax lesion image asked about pneumothorax (only 7, 7 and 4 lesion-free ones), so the selection rule could not see the problem.

Stage 3 turns weak detection into none. Stage 3 shifts chest X-ray answers towards "No" in general: on validation, named findings detected fall from 63% to 56%, erased-lesion claims from 16% to 9% and healthy-image claims from 24% to 10%. With almost no positive pneumothorax examples, its weak detection drops to zero at every lesion size. The share of "No" targets alone does not explain this: stage 2 trained on 35 of 35 "No" targets and still detected 40%.

The model registers something. Asked "Is there an abnormality?", MedScope-4B says yes on 35 of 42 pneumothorax images and on 28 of 48 SIIM images without pneumothorax (Fisher $p = 0.012$); the base model separates them more (38/42 vs 23/48, $p < 0.001$). The final model more often reports a generic abnormality on positive than on negative images; this does not identify which cue it uses.

Lessons. The evidence is consistent with low positive coverage of a rare critical finding, amplified by a final stage that shifts answers towards "No"; it is a post-hoc diagnosis, not a controlled demonstration, and a repair would need a pre-registered coverage experiment with a new locked test cohort. The practical lesson stands regardless: validation sets need positives of every critical finding, and selection rules need a per-finding sensitivity floor, or a conservative final stage can remove a finding entirely while every aggregate metric improves. Coverage may not be the whole story: the gap to gemini-4-argon even at full resolution is consistent with the frozen vision encoder limiting subtle findings.

7 Reader study

The benchmark's labels come from datasets, and the erasures are checked only automatically. Physicians therefore read a blinded set of test images to check three assumptions: that an erased finding is no longer visible, that erased images are not trivially recognisable as edited, and that the control edit is as recognisable as the erasure, so that noticing an edit does not tell a reader which kind it is. [Results pending; this section gives the design and the analysis fixed before any answer was seen.]

Images. 128 images from the main test, each from a different case and stratified by source: 32 unmodified images with a lesion, 32 lesion-erased images (the LaMa erasures the models were tested on), 32 control-edited images (the same shape erased elsewhere with the same tool, lesion intact) and 32 unmodified lesion-free images. Lesion-free images are included so that "no lesion" is not a cue for "edited", and control-edited images so that readers rate the control edit as well as the erasure. Only yes/no presence questions are used. Every image is re-saved the same way (PNG, no metadata) under an opaque name.

Procedure. Readers use a static web page that holds no answer key, in their own random order, with no time limit and no other information. For each image they answer (1) whether the image has been digitally edited (definitely real, probably real, probably edited, definitely edited) and (2) the case's clinical question (yes, no, cannot tell). They see no results, no paper and no image counts by kind until their answers are submitted; a session takes about 1.5 to 2 hours. Every answer file names the pack it was answered on, and only files from the final pack are scored.

Pre-specified analysis. Per reader: the share of erased, control-edited and unmodified images rated probably or definitely edited; the AUC for separating edited from unmodified images on the four-level rating, and the AUC for separating erased from control-edited images (0.5 means the two edits are equally recognisable); the share answering "yes" to the clinical question on each image kind; agreement between readers (Cohen's κ on edited yes/no); and results by source. The pre-specified criterion is the residual-finding rate (erased images on which a reader still answers "yes") with its exact 95% confidence interval, read against an acceptable rate of 10%; with 32 erased images, even 0 residual findings gives an upper

bound of about 11%, so this study can rule out gross failure of the erasures, not small residual rates. Edit visibility is reported separately as the AUC, with the zero-shot frontier models' 0.70–0.75 as the reference.

Readers. [Number, specialty, years of experience.] **Results.** [Pending.]

8 Discussion

8.1 Principal findings

Seeing a lesion and reporting only what is seen are separable abilities. Nine frontier models see real lesions as well as a 4B model, yet claim about half of the lesions after they have been erased, under the benchmark prompt, under a cautious prompt, with solved examples in context, at their highest reasoning and image-processing settings, with their own free reply, and with reworded questions. A small model acquires a disposition to report findings only when it sees them from automatically built counterfactual views, and the ablations support the counterfactual recipe, rather than the data, the answer format or the share of "No" answers, as what produces it: the erasures give evidence sensitivity, control edits give real-lesion accuracy and reduce the "edited means absent" shortcut, and lesion insertions help on one test set and not on another. Two pre-registered checks bound the claim: matching the control edit to the erasure was not shown to add anything, and removal-based preference training on the same cases reaches the same evidence sensitivity on new data while keeping more sensitivity and claiming more erased lesions. What the data support is therefore that a small model can be taught, from automatically built erased and edited images and without human annotation, to stop reporting what it does not see; not that one particular construction of the control is required, nor that supervised counterfactual training is the only way. They do not exclude every residual shortcut. The effect holds on every seed, on a second base model, and on inpainting tools never used in training. Because the views come from public masks and automatic checks, the recipe can be applied to any dataset with segmentation labels whose masks permit a clinically valid erasure, which must be checked for each task.

8.2 What it is not

The gain is in evidence sensitivity, not in perception. At new centres the frontier models detect more real lesions (CCR 0.95–0.97 vs 0.73), and the two abilities pull net utility in opposite directions: frontier models behave as high-sensitivity, high-hallucination readers, MedScope-4B as an evidence-gated reader that misses more unfamiliar lesions but rarely reports absent ones. Which is preferable depends on the clinical cost of each error, which our cost matrix fixes only by assumption. The fresh external test shows the danger of the evidence-gated reader at its worst: where perception or training coverage is weak, conservatism becomes a complete miss of a critical finding.

8.3 Relation to prior work

Hallucination benchmarks and training-free detectors [9, 11, 12, 13, 14, 16, 17] measure or flag unsupported claims; we show that the same counterfactual idea, applied to the image with an edit-matched control, can train the behaviour into a model. Compared with natural-image counterfactual VQA [18], control edits reduce the shortcut of reacting to edits, and our erasures-only arm shows that without them a model partly learns that shortcut; whether the control matches the erasure's shape and tool did not measurably matter. Compared with removal-based preference optimisation [10], the adapted baseline reaches the same evidence sensitivity on new data, so the two lines of work should be read as complementary: the recipe's distinctive effect is fewer claims of erased findings, the baseline's is sensitivity. Compared with tool-using agents trained by reinforcement learning [22, 23, 24, 25], label-derived trajectories and rejection sampling were enough to teach tool use (first zoom on the lesion in 93% of lesion views).

8.4 Implications

For developers, a model's answers to presupposing questions should be tested on images with the finding removed, not only on images where it is present. For evaluators, aggregate accuracy and even aggregate hallucination rates can hide a complete miss of one critical finding; per-finding sensitivity floors belong in model selection. For clinical use, MedScope-4B is a research model, not a medical device, and nothing here supports unsupervised use.

8.5 Limitations

- **Rare critical findings.** A finding almost absent from training and absent from validation was lost entirely by the most conservative stage (Section 6).
- **Perception.** The vision encoder is frozen; subtle findings and the smallest lesions (ESG 0.55 in the lowest quartile) remain weakest.
- **Erasure as ground truth.** Diffuse findings are erased less completely (QaTa-COV19 ESG 0.05), erasure can leave secondary signs such as mass effect or oedema, and a second, unmasked lesion (Figure 2, breast ultrasound) makes the expected "absent" answer wrong for every model. The reader study tests how often an erased finding remains visible.
- **Cross-dataset provenance.** Splits are disjoint by patient or near-copy group within each dataset, and no image of one split or cohort recurs in another at the pixel level (Section 4.1). Datasets that re-identify the same hospital's patients under new identifiers (SIIM-ACR and ChestX-Det both draw on NIH ChestX-ray14) could still place different images of one patient in training and in the fresh external set; this cannot be excluded without the source identifiers, which SIIM-ACR does not carry, and it would make the fresh external result for pneumothorax, where the model failed, if anything optimistic rather than pessimistic.
- **Edit traces.** A trained detector finds the edits (AUC 0.83). The advantage holds on clean-looking erasures and unseen tools, but a residual contribution of edit traces cannot be excluded.
- **Reference standard.** Labels and masks come from public datasets without re-reading, and the "absent" answer expected after erasure has not yet been validated by readers (Section 7); the model-consensus check of lesion-free labels is a sensitivity analysis, not clinical adjudication.
- **ESG is not immune to artefacts.** It is zero for policies that respond to both edits alike, not for a detector sensitive to anatomy-dependent traces of erasure; the frozen encoder carries a weak edit signal (probe AUC 0.60), and Section 5.4 bounds but does not eliminate this.
- **Dependence between cases.** Cases are images and several can come from one patient; patient-level resampling changes one secondary comparison and no primary one.
- **Lesion insertions.** The pre-registered recipe includes them; a single-seed factorial finds no benefit on the main test and a benefit on the fresh external set. The recipe was kept as registered.
- **Single-seed mechanism checks.** The factorial, the unmatched-control arm and the preference baseline are one seed each (the baseline's further seeds are pre-registered); differences near the recipe's seed-to-seed spread (about 0.02–0.03) are flagged as such in the released reports.
- **Utility under a response bias.** On a finding no arm detects (SIIM pneumothorax), an arm that answers "present" indiscriminately scores higher net utility than one that answers "No", because the cost matrix charges a missed critical finding most. Net utility must be read with sensitivity and erased-lesion claims beside it.
- **Same construction for training and test.** MedScope trained on views built like the test views; the frontier models saw none. The ablation trained on the same data without counterfactual views hallucinates at the frontier level, which supports the views as the main contributor under this training design, and solved examples in context did not close the gap.
- **Comparator configuration.** Frontier models were evaluated through public APIs on fixed dates, open models with the shared agent prompt or one generic plain prompt. The flagships were also tried with a cautious prompt, solved examples, their highest reasoning and image-processing settings and their own free reply, none of which closed the gap; but model-specific instructions were not tuned for any comparator, every model received images of at most 768 pixels, and API behaviour may change.

- **Scope.** Yes/no and short-answer questions on single 2D images; no report generation, no 3D volumes, no prospective or clinical evaluation, and limited demographic information in the source datasets.
- **No human annotation.** All checks are automated or model-based until the reader study is complete.

8.6 Future work

Three directions follow directly: per-finding coverage and sensitivity floors in data and selection; unfreezing or replacing the vision encoder so that evidence sensitivity is combined with frontier-level perception; and extending the views to report generation, where a removed finding should disappear from the report while the rest stays.

9 Conclusion

Erasing the lesion, with an edit elsewhere as the control, turns "does the answer follow the image?" into a property that can be measured and trained without human annotation. Trained on such views and selected by a pre-registered rule, a 4B agent claims erased findings at 4–6 times lower rates than nine frontier models, with no significant difference in accuracy on real lesions on the internal test, robustly to prompting, question wording and unseen inpainting tools, while keeping aggregate VQA accuracy. Pre-registered checks show that the control edit need not match the erasure, and that removal-based preference training reaches the same evidence sensitivity with a different error profile; the contribution is the measurement and the demonstration, not a unique training principle. Its perception is weaker than the frontier models' at new centres, and it is only as good as the findings its training data covers: a pre-registered fresh test exposed a complete miss of pneumothorax. Data coverage and per-finding selection safeguards are the priority hypotheses for repair; a successful repair remains to be shown on new held-out data.

Declarations

Data and code availability. Code, view-building recipes, notebooks, the pre-registration with its git history, and the results log: `https://github.com/bhaskargurram-ai/medscope`. LoRA adapters and MedScope-Bench (view manifests and per-view transcripts of every evaluation run) on Hugging Face (`bhaskar-ai/MedScope-4B`; `datasets/bhaskar-ai/MedScope-Bench`), public on acceptance. All images come from public de-identified datasets under their own licences and are not redistributed.

Funding. This research received no external funding. **Competing interests.** The author declares no competing interests. **Ethics.** The model study used only public, de-identified datasets under their published licences; no new patient data were collected. [Reader study: participant consent and local ethics position.] **Author contributions (CRediT).** Bhaskar Gurram: conceptualisation, methodology, software, investigation, formal analysis, data curation, writing (original draft, review and editing), visualisation. [Reader-study physicians: acknowledged or credited as agreed.] **Declaration of generative AI use.** [To be completed according to the target journal's policy.]

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A Prompts

Agent prompt (every model, all main results):

```
You are MedScope, a careful medical imaging assistant used for research. You look at a medical
    image and answer a question about it from what the image actually shows: you report a finding
     only when you can see it, and you say so when it is not there.
On each turn, think briefly in plain text, then write exactly ONE action as a tagged JSON object
    and stop. After a tool call, stop and wait: you will be shown the result before you continue.
Example of a tool turn:
The lesion may be small, so I will look closer at the upper left. <tool>{"name": "zoom", "box":
    [100, 150, 450, 500]}</tool>
Example of a final turn:
The mass is clearly visible. <final>{"action": "answer", "answer": "Yes", "box": [180, 220, 320,
    360], "confidence": 0.85}</final>
Tools (at most 3 calls per case; boxes are integers 0-1000 on the ORIGINAL full image, as [x1, y1
    , x2, y2]):
<tool>{"name": "zoom", "box": [x1, y1, x2, y2]}</tool> look closer at a region
<tool>{"name": "window", "center": 128, "width": 100}</tool> adjust brightness/contrast (0-255
    scale)
Final actions (exactly one, then stop):
<final>{"action": "answer", "answer": "...", "box": [x1, y1, x2, y2], "confidence": 0.0-1.0}</
    final> when the image lets you answer. If the finding is visible, the box marks it on the
    original image. If the finding asked about is not there, say so and leave out the box: "No"
    to "Is there ...?", "Yes" to "Is ... healthy?" or "Is ... normal?", "No abnormality" to an
    open question.
<final>{"action": "clarify", "question": "..."}</final> when the image is unusable (too blurred,
    noisy, low resolution, or not the anatomy asked about).
<final>{"action": "refer", "reason": "..."}</final> when the image is usable but you cannot
    decide either way.
Keep answers short, using the wording a radiologist would use (for example "Yes", "No", "Liver",
    "Left lung"). Never report a finding you cannot see.
```

Plain prompt (open models, plain-prompt runs; one generic prompt for every open model, applied through each model's own chat template): "You are a medical imaging assistant. Look at the image and answer the question. Keep the answer short." One reply, no tools. **Cautious prompt** (Table 12): the agent prompt followed by "Important: a question may name a finding that is not in the image. Do not assume the question is correct. Before answering, check carefully whether the finding is actually visible. If you cannot clearly see it, say it is not there ("No"). Reporting a finding that is not visible is a serious error." **Few-shot condition** (Table 12): the cautious prompt plus four solved training examples before each case (a lesion; the same image with it erased; an edit elsewhere with the lesion intact; a healthy image). Example images never appear in any test set.

B Reproducibility

- **Hardware.** One NVIDIA A100 40 GB per training stage (about 4 hours each); one local model on the main test, about 1 hour. Frontier models run from a CPU runtime.
- **Determinism.** Greedy decoding for open models; the 16,000-view stage-1 draw is fixed across seeds; seeds change LoRA initialisation, data order and trajectory sampling.
- **Re-scoring.** Every table is regenerated from stored per-view transcripts by one script; the notebook behind each number is named in the results log. The factorial ablation, the encoder probe and the patient-level bootstrap were added after an independent review of draft v0.7 (notebook 29); the review and the changes it led to are in the repository.
- **Pre-registration.** `docs/PREREGISTRATION_v4.md`, committed before any v4 data existed; its git history records the rule, both amendments and both post-hoc notes. The v5 and v6 plans fix the mechanism checks of Section 3.7 before their arms were trained, and `scripts/v5_confirm.py` and `v6_confirm.py` apply their rules verbatim.
- **Negative results from development.** Plain GRPO from a stage-1 agent gave reward differences in only 20% of sampled groups, so group-relative advantages vanished; dynamic sampling gained one validation case. Merging a grounding-only adapter into the agent removed evidence sensitivity (abstention gap 0.90

$\rightarrow$ 0.00) while keeping grounding. An adapter trained only to answer with boxes claimed the finding on every erased and every lesion-free image.

C CLAIM 2024 checklist (summary)

Item	Where addressed
Study design (retrospective); intended use (research)	Sections 1, 4.1; Discussion
Data sources, inclusion and exclusion	Section 4.1, Tables 3–4
De-identification	Public de-identified sources; no new patient data
Reference standard and its limits	Sections 3.2, 8.5; reader study (Section 7)
Partitions and disjointness	Patient or near-copy-group splits; test protection (Section 4.1)
Model, initialisation, training details	Section 3.5; Appendix B
Selection of the final model	Section 3.6 (pre-registered)
Metrics, uncertainty, statistical tests	Sections 3.8, 3.9
Robustness and sensitivity analyses	Sections 5.3–5.5
Internal and external testing	Sections 5.1, 5.6
Failure analysis	Section 6
Availability of code, model, data	Declarations